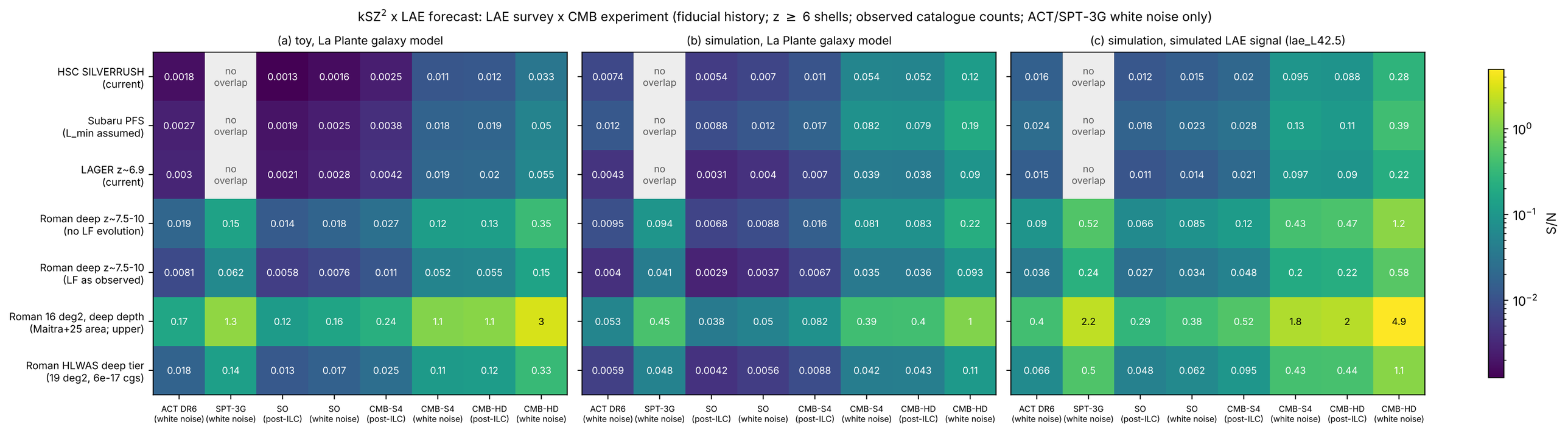

kSZ² x LAE forecast: LAE survey x CMB experiment (fiducial history; z ≥ 6 shells; observed catalogue counts; ACT/SPT-3G white noise only)
(a) toy, La Plante galaxy model
(b) simulation, La Plante galaxy model
(c) simulation, simulated LAE signal (lae_L42.5)
HSC SILVERRUSH (current)
Subaru PFS (L_min assumed)
LAGER z~6.9 (current)
Roman deep z~7.5-10 (no LF evolution)
Roman deep z~7.5-10 (LF as observed)
Roman 16 deg2, deep depth (Maitra+25 area; upper)
Roman HLWAS deep tier (19 deg2, 6e-17 cgs)
0.0018 no overlap 0.0013 0.0016 0.0025 0.011 0.012 0.033
0.0027 no overlap 0.0019 0.0025 0.0038 0.018 0.019 0.05
0.003 no overlap 0.0021 0.0028 0.0042 0.019 0.02 0.055
0.019 0.15 0.014 0.018 0.027 0.12 0.13 0.35
0.0081 0.062 0.0058 0.0076 0.011 0.052 0.055 0.15
0.17 1.3 0.12 0.16 0.24 1.1 1.1 3
0.018 0.14 0.013 0.017 0.025 0.11 0.12 0.33
0.0074 no overlap 0.0054 0.007 0.011 0.054 0.052 0.12
0.012 no overlap 0.0088 0.012 0.017 0.082 0.079 0.19
0.0043 no overlap 0.0031 0.004 0.007 0.039 0.038 0.09
0.0095 0.094 0.0068 0.0088 0.016 0.081 0.083 0.22
0.004 0.041 0.0029 0.0037 0.0067 0.035 0.036 0.093
0.053 0.45 0.038 0.05 0.082 0.39 0.4 1
0.0059 0.048 0.0042 0.0056 0.0088 0.042 0.043 0.11
0.016 no overlap 0.012 0.015 0.02 0.095 0.088 0.28
0.024 no overlap 0.018 0.023 0.028 0.13 0.11 0.39
0.015 no overlap 0.011 0.014 0.021 0.097 0.09 0.22
0.09 0.52 0.066 0.085 0.12 0.43 0.47 1.2
0.036 0.24 0.027 0.034 0.048 0.2 0.22 0.58
0.4 2.2 0.29 0.38 0.52 1.8 2 4.9
0.066 0.5 0.048 0.062 0.095 0.43 0.44 1.1
ACT DR6 (white noise)
SPT-3G (white noise)
SO (post-ILC)
SO (white noise)
CMB-S4 (post-ILC)
CMB-S4 (white noise)
CMB-HD (post-ILC)
CMB-HD (white noise)
10^0
10^-1
10^-2
N/S